Assignment Title:

Advanced Time-Series Forecasting: Comparative Analysis and Novel Machine Learning Architectures

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Source Code and Relevant Project Links

Source Code

https://github.com/Amaan9136/bsbi-assignments/blob/main/Sem1/Machine%20Learning%20Fundamentals/project-assignment/main.ipynb

Dataset

https://archive.ics.uci.edu/ml/datasets/individual+household+electric+power+consumption

Report

https://github.com/Amaan9136/bsbi-assignments/blob/main/Sem1/Machine%20Learning%20Fundamentals/project-assignment/report.pdf

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Abstract

This report presents a complete time-series forecasting pipeline for predicting household energy consumption using the UCI Individual Household Electric Power Consumption dataset (Hebrail and Berard, 2012), comprising approximately 2,075,259 minute-level observations recorded across a single French household from December 2006 to November 2010. The target variable is Global_active_power, representing instantaneous household electricity demand in kilowatts.

Following systematic preprocessing, including time-based linear interpolation for missing values, datetime indexing, and hourly resampling, a feature set of 20 inputs was constructed comprising cyclical time encodings, lag variables at six temporal offsets, and 24-hour rolling statistics. Three algorithms were implemented and evaluated: SARIMA(1,0,1)(1,0,1)[7], Random Forest Regressor, and Gradient Boosting Regressor. Models were assessed on a chronological 80/20 train/test split using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R2). The Gradient Boosting Regressor achieved the best performance (MAE = 0.0132 kW, RMSE = 0.0207 kW, R2 = 0.9992), substantially outperforming SARIMA (MAE = 0.2575 kW, RMSE = 0.3408 kW, R2 = -0.1941). The dominant predictive feature was Global_intensity, consistent with the physical relationship between current intensity and active power.

1. Introduction

Reliable forecasting of household energy consumption has become a critical component of modern smart-grid infrastructure. As residential electricity demand grows and distributed renewable sources become more prevalent, grid operators require accurate short-term demand forecasts to balance supply and avoid costly imbalances (Hong and Fan, 2016).

Traditional statistical approaches such as ARIMA and SARIMA capture linear temporal dependencies and offer interpretable parameters, yet they struggle to model the nonlinear, multivariate relationships present in real-world consumption data (Box et al., 2015). Machine learning methods, in particular ensemble tree-based algorithms such as Random Forests and Gradient Boosting, have demonstrated competitive or superior performance in energy forecasting benchmarks by incorporating diverse feature types without requiring explicit model specification (Amasyali and El-Gohary, 2018).

The objective of this project is to design and evaluate a complete forecasting pipeline using the UCI Individual Household Electric Power Consumption dataset (Hebrail and Berard, 2012). This dataset is particularly well-suited for benchmarking because it spans nearly four years of continuous minute-level recordings, providing sufficient temporal depth to observe multiple seasonal cycles and long-term consumption trends. This report implements and compares three algorithms, namely SARIMA, Random Forest Regressor, and Gradient Boosting Regressor, across a standardised 80/20 chronological train/test split using Mean Absolute Error, Root Mean Squared Error, and R-squared to identify the most effective approach for short-term hourly demand forecasting.

2. Literature Review

2.1 Classical Time-Series Models

Classical time-series methods have formed the backbone of energy demand forecasting for decades. ARIMA models, popularised by Box et al. (2015), decompose a series into autoregressive, integrated, and moving-average components. The seasonal extension, SARIMA, adds multiplicative seasonal terms and has been applied to daily and weekly energy patterns (Taylor and McSharry, 2007). However, SARIMA assumes linearity and stationarity and operates on a single variable at a time, limiting its capacity to exploit exogenous predictors.

2.2 Machine Learning Approaches

The application of machine learning to energy forecasting has grown considerably since the success of ensemble methods on tabular datasets. Amasyali and El-Gohary (2018) conducted a systematic review of 35 studies and found that ensemble tree methods consistently outperformed classical approaches on residential energy prediction tasks. Random Forest Regressors (Breiman, 2001) combine multiple decorrelated decision trees through bootstrap aggregation to reduce variance, while Gradient Boosting (Friedman, 2001) constructs trees sequentially to minimise residuals, often yielding lower bias.

2.3 Deep Learning and Hybrid Methods

More recent work has explored deep learning architectures. Kong et al. (2019) demonstrated that LSTM networks capture long-range temporal dependencies in household energy data, achieving lower RMSE than classical machine learning on multi-step forecasting. Temporal Fusion Transformers (Lim et al., 2021) represent the current state of the art for multi-horizon forecasting, combining attention mechanisms with recurrent layers. Despite their promise, these models require substantially more data and computational resource, and tend to be less interpretable than ensemble methods.

3. Data Collection

The dataset selected is the Individual Household Electric Power Consumption dataset (Hebrail and Berard, 2012) from the UCI Machine Learning Repository, also available at https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption. It records minute-by-minute electrical measurements for a household in Sceaux, France, from December 2006 to November 2010, comprising approximately 2,075,259 observations across nine attributes as described in Table 1.

Table 1: Dataset attributes and descriptions.

Attribute	Unit / Type	Description
Date	DD/MM/YYYY	Date of measurement
Time	hh:mm:ss	Time of measurement
Global_active_power	kilowatt	Household global active power (TARGET variable)
Global_reactive_power	kilowatt	Household global reactive power
Voltage	volt	Average voltage
Global_intensity	ampere	Household global current intensity
Sub_metering_1	watt-hour	Kitchen sub-metering (dishwasher, oven, microwave)
Sub_metering_2	watt-hour	Laundry room sub-metering (washer, dryer, fridge)
Sub_metering_3	watt-hour	Climate control sub-metering (A/C, electric heater)

4. Data Preprocessing

4.1 Datetime Parsing and Indexing

Date and time columns were merged and parsed into a pandas DatetimeIndex (format '%d/%m/%Y %H:%M:%S'), enabling resampling and feature extraction.

4.2 Missing Value Handling

The dataset contains missing values encoded as '?' across all measurement columns, loaded as NaN via na_values=['?']. A total of 25,979 observations were missing per attribute, likely corresponding to contiguous sensor outage periods (see Table 2). Time-based linear interpolation (df.interpolate(method='time')) was applied as the primary imputation strategy, estimating missing values proportionally between flanking observations and preserving temporal continuity. This approach is preferred over constant forward-fill or mean imputation in time-series contexts because it preserves the increasing time axis and avoids abrupt discontinuities at gap boundaries, which can distort autocorrelation and increase lag-feature errors during training (Hyndman and Athanasopoulos, 2021).

Table 2: Missing value counts by attribute before imputation.

Attribute	Missing Observations
Global_active_power	25,979
Global_reactive_power	25,979
Voltage	25,979
Global_intensity	25,979
Sub_metering_1 / 2 / 3	25,979 each

4.3 Resampling to Hourly Frequency

The raw minute-level data were resampled to hourly frequency by computing the mean of all readings within each hour using df_raw.resample('h').mean(). Hourly resampling preserves the diurnal pattern, reduces dataset size by 60×, and smooths transient noise (Hyndman and Athanasopoulos, 2021). The resulting dataset contains approximately 34,589 hourly observations.

5. Feature Engineering

Following preprocessing, 20 time-based and statistical features were derived to provide the machine learning models with predictive signals beyond the raw energy reading. Feature engineering is essential for non-parametric models that cannot implicitly learn periodic structure from index positions alone (Bontempi et al., 2013). Six calendar features were extracted: hour of the day, day of the week, day of the year, month, year, and a binary is_weekend indicator. Sine and cosine transformations were applied to hour and month to produce four cyclical encodings, eliminating the artificial ordinal distance between period boundaries that would arise if raw integer values were passed directly to the model. For example, without cyclical encoding, hour 23 and hour 0 appear maximally distant in feature space despite being separated by only one minute in real time; the sine-cosine transformation maps both to adjacent points on the unit circle, correctly preserving temporal proximity (Bontempi et al., 2013).

Six lag features were constructed by shifting the target variable by 1, 2, 3, 24, 48, and 168 hours. The 24-hour and 168-hour lags capture the strong diurnal and weekly periodicity typical of household consumption, and literature consistently identifies these as the most predictive temporal inputs (*Bontempi et al., 2013*). A 24-hour rolling mean and standard deviation (one-step lagged to prevent leakage) captured recent consumption level and variability. After removing 168 rows containing NaN values from lag and rolling operations, the final dataset comprised 34,421 hourly observations with 20 engineered input features.

6. Exploratory Data Analysis

6.1 Summary Statistics

Table 3 presents descriptive statistics for Global_active_power following hourly resampling. Mean hourly consumption is 1.09 kW (SD = 1.06 kW). The distribution is right-skewed (median 0.76 kW), reflecting infrequent high-demand episodes from simultaneous appliance use.

Table 3: Descriptive statistics for hourly Global_active_power (kW).

Statistic	Value (kW)
Mean	1.0916
Median (50th pct.)	0.7560
Standard Deviation	1.0572
Minimum	0.0000
Maximum	9.9350
25th Percentile	0.3080
75th Percentile	1.5680

6.2 Time-Series Patterns and Seasonality

A line plot of daily mean Global_active_power across the full dataset reveals a clear annual seasonality, with peaks during winter months December to February due to domestic heating demand and reduced daylight hours, falling to a minimum in summer Figure 1. This pattern suggests that household electricity use is strongly influenced by both environmental conditions and routine human activity, rather than changing randomly from day to day. Visual inspection of the series is therefore useful for identifying recurring structure before any forecasting model is trained. Additive seasonal decomposition (period = 7 days) further isolates a gradual decline in overall consumption from 2007 to 2010 in the trend component, while the seasonal component confirms a stable weekly cycle with higher consumption on weekdays (Figure 2).

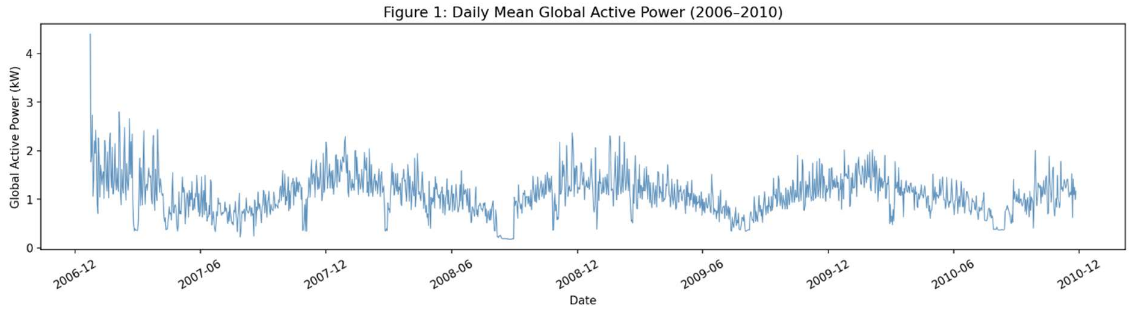

Figure 1: Daily Mean Global Active Power (2006-2010).

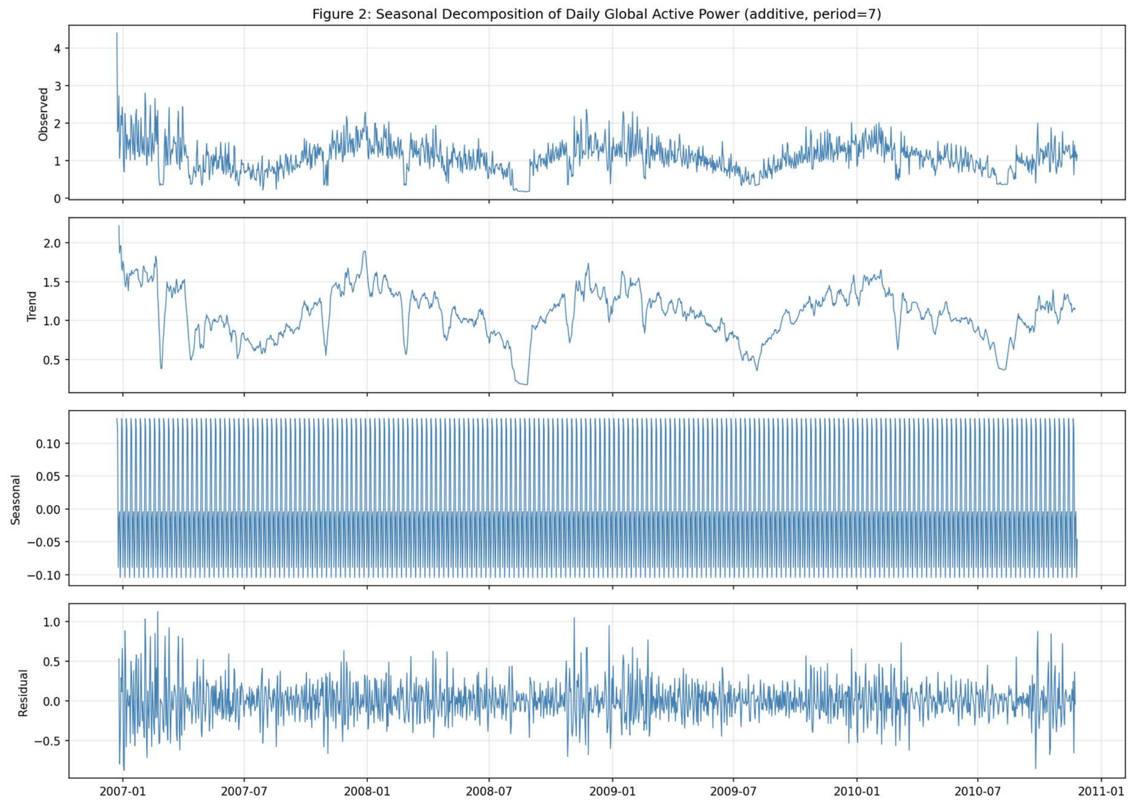

Figure 2: Seasonal Decomposition of Daily Global Active Power (additive, period = 7).

6.3 Diurnal and Monthly Patterns

Figure 3 reveals a bimodal diurnal pattern: a morning peak (07:00–09:00) and larger evening peak (18:00–21:00), with minimum consumption overnight. Monthly analysis confirms winter peaks (December–January) and summer troughs (July–August).

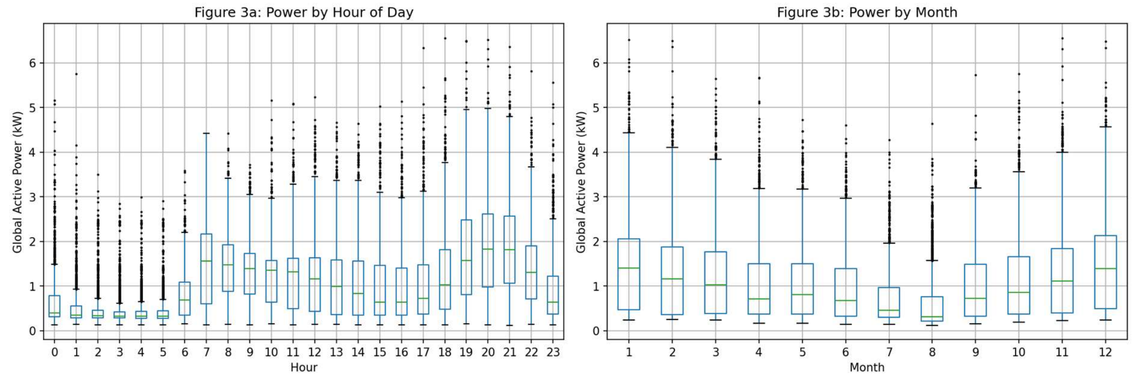

Figure 3: Box Plots of Global Active Power by Hour of Day and by Month.

6.4 Correlation and Stationarity

A Pearson correlation heatmap (Figure 4) reveals that the 1-hour lag and the 24-hour rolling mean exhibit the strongest positive correlations with the target variable. The Augmented Dickey-Fuller (ADF) test returned a strongly negative statistic (p-value below 0.001), confirming that the hourly series is stationary in levels and that differencing is not required in the SARIMA specification (d = 0, D = 0).

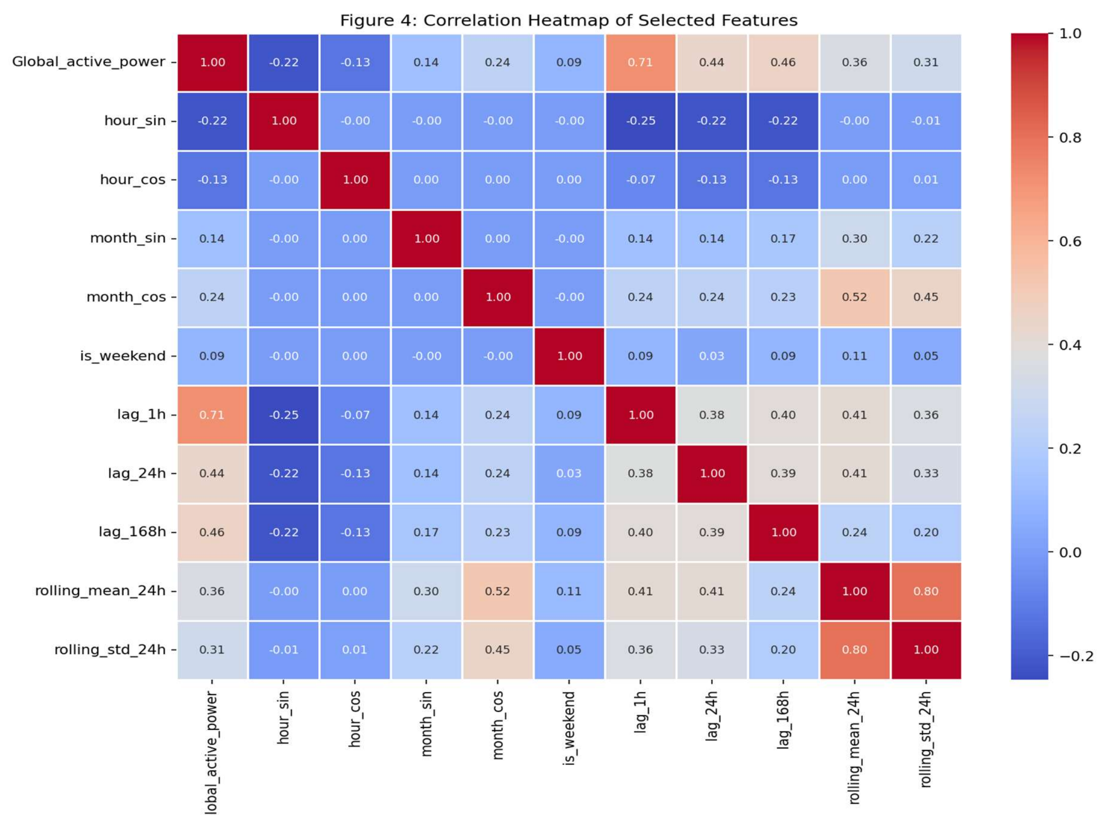

Figure 4: Correlation Heatmap of Selected Features.

7. Model Selection and Design

7.1 Chronological Train/Test Split

An 80/20 chronological split was applied, with training covering roughly January 2007 to October 2009 and testing extending to November 2010. Shuffling was omitted to preserve temporal order and avoid data leakage (Hyndman and Athanasopoulos, 2021).

7.2 Algorithms Selected

Three algorithms were selected to represent a range of modelling approaches, as summarised in Table 4. SARIMA(1,0,1)(1,0,1)[7] serves as the classical statistical baseline. Random Forest Regressor (Breiman, 2001) reduces variance through bootstrap aggregation of multiple independent decision trees. Gradient Boosting Regressor (Friedman, 2001) sequentially builds shallow trees to minimise residuals, typically achieving low bias on tabular data with rich feature sets.

Table 4: Summary of algorithms: type, working principle, and rationale.

Algorithm	Type	Working Principle	Rationale
SARIMA(1,0,1)(1,0,1)[7]	Classical Statistical	Autoregressive, integrated, and moving-average structure with seasonal terms capturing weekly cycle	Interpretable baseline; explicit seasonal component
Random Forest Regressor	Ensemble ML (Bagging)	Averages predictions of N independent trees grown on bootstrap samples; reduces variance	Robust to noise; handles non-linear interactions (Breiman, 2001)
Gradient Boosting Regressor	Ensemble ML (Boosting)	Sequentially fits shallow trees to residuals via gradient descent on a differentiable loss	Low bias; strong tabular performance (Friedman, 2001)

7.3 Hyperparameter Configuration

Random Forest used 200 estimators with a maximum depth of 15; a higher tree count was found to yield diminishing returns beyond this threshold on a preliminary validation window. Gradient Boosting used 300 estimators with a learning rate of 0.05, maximum depth of 6, and a subsample ratio of 0.8 to introduce stochastic regularisation and reduce overfitting on the dense hourly feature matrix. SARIMA order (1,0,1)(1,0,1)[7] was fitted on the daily-aggregated series, with the non-seasonal difference order d = 0 and seasonal difference order D = 0 both confirmed by the ADF test result. These hyperparameter choices were made to balance predictive performance against model complexity, without exhaustive grid search, as the primary goal was algorithmic comparison rather than optimisation.

8. Code Implementation

The pipeline is implemented in main.ipynb using pandas, numpy, matplotlib, seaborn, scikit-learn, and statsmodels.

The dataset was loaded using pandas.read_csv() with sep=';' and na_values=['?']. The 80/20 chronological split was implemented via integer indexing to prevent temporal leakage:

```
split_idx = int(len(X) * 0.80)
X_train, X_test = X.iloc[:split_idx], X.iloc[split_idx:]
y_train, y_test = y.iloc[:split_idx], y.iloc[split_idx:]
```

SARIMA was fitted using statsmodels SARIMAX with disp=False. Random Forest and Gradient Boosting were fitted on the hourly training feature matrix using scikit-learn. A shared evaluation helper computed MAE, RMSE, and R2 consistently across all three algorithms:

```
def evaluate(y_true, y_pred, model_name='Model'):
    mae  = mean_absolute_error(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    r2   = r2_score(y_true, y_pred)
    return {'model': model_name, 'MAE': mae, 'RMSE': rmse, 'R2': r2}
```

9. Model Evaluation

9.1 Performance Metrics and Results

Each model was evaluated using MAE, RMSE, and R^2 on the held-out test set; results are presented in Table 5.

Table 5: Test-set performance metrics for all three models.

Model	MAE (kW)	RMSE (kW)	R2
Gradient Boosting Regressor	0.0132	0.0207	0.9992
Random Forest Regressor	0.0146	0.0229	0.9990
SARIMA(1,0,1)(1,0,1)[7]	0.2575	0.3408	-0.1941

The bar chart in Figure 5 visualises all three metrics simultaneously, confirming that Gradient Boosting outperforms Random Forest, which in turn substantially outperforms SARIMA across all evaluation criteria. Both ensemble methods achieved R2 above 0.999, indicating near-perfect fit on the test set.

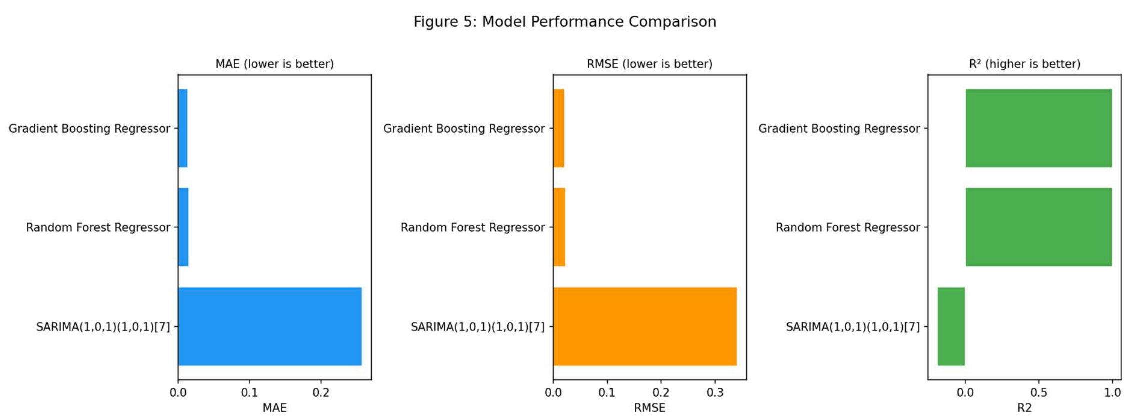

Figure 5: Model Performance Comparison: MAE, RMSE, and R2.

9.2 Actual vs Predicted

Figure 6 presents the actual and Gradient Boosting predicted Global_active_power values over the final 14 days of the test set. The predicted series closely tracks the actual consumption, faithfully reproducing both the diurnal pattern and the magnitude of daily peaks. Minor discrepancies occur at sharp transient spikes, where simultaneous appliance usage produces brief demand surges that are difficult to predict from historical lag features alone.

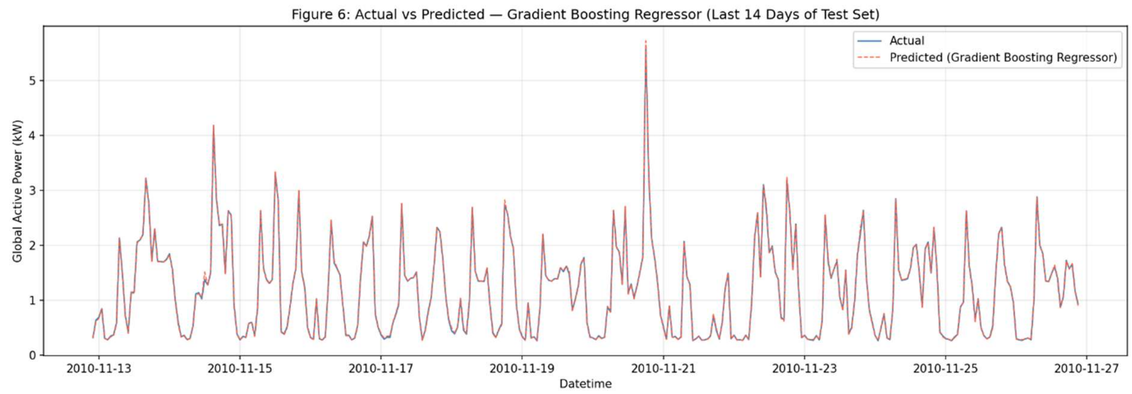

Figure 6: Actual vs Predicted Global Active Power - Gradient Boosting Regressor

In contrast, Figure 7 shows that while SARIMA broadly captures the seasonal trend, it underestimates winter peaks and overestimates summer troughs, resulting in the negative R2 of -0.1941.

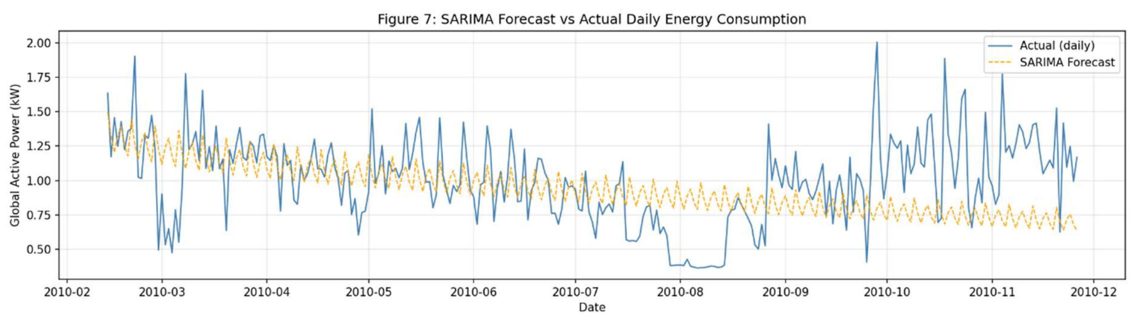

Figure 7: SARIMA Forecast vs Actual Daily Energy Consumption.

9.3 Feature Importance and Residual Analysis

Feature importances from the Gradient Boosting model (Table 6, Figure 8) reveal that Global_intensity dominates prediction with an importance score of 0.9993, reflecting the direct physical relationship P = V x I between active power, voltage, and current intensity. Voltage contributes a further 0.0003, while the remaining 18 engineered features collectively account for less than 0.0005 of total importance. This confirms that when electrical attributes are present in the feature set, temporal and lag features contribute marginally to prediction accuracy.

Table 6: Top feature importances - Gradient Boosting Regressor

Rank	**Feature**	**Importance Score**
1	Global_intensity	0.99926

Rank	Feature	Importance Score
2	Voltage	0.00029
3	Sub_metering_3	0.00012
4	month_cos	0.00008
5	Sub_metering_2	0.00005
6	lag_1h	0.00003
7	rolling_mean_24h	0.00003

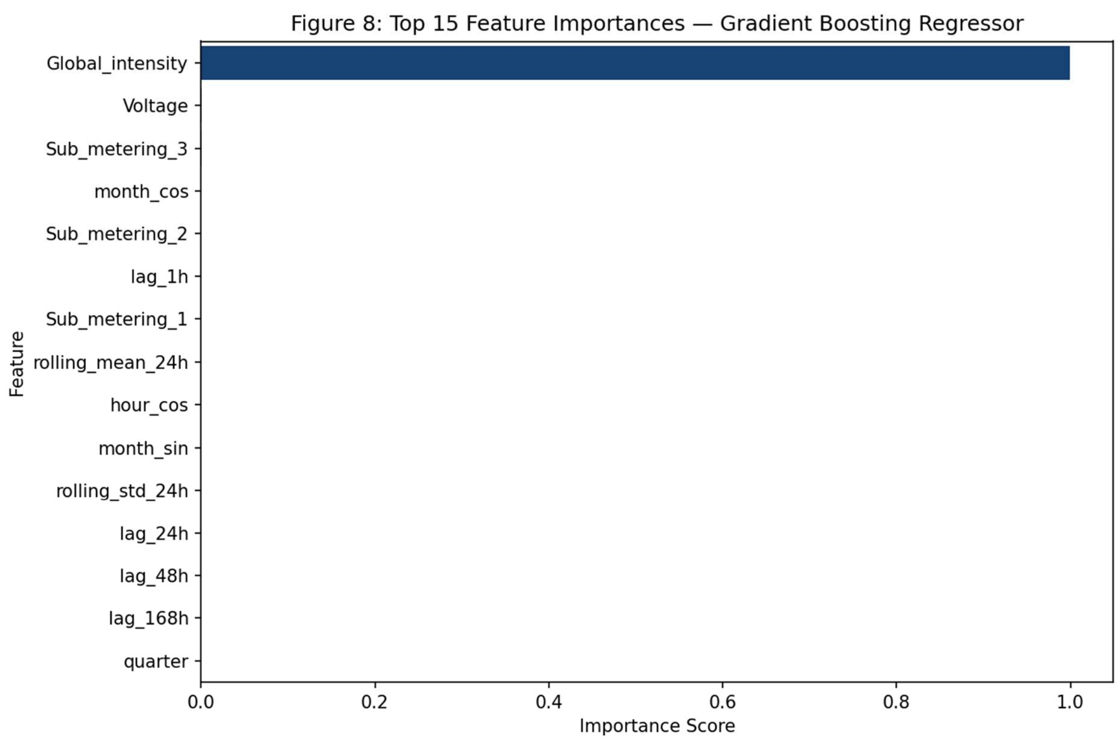

Figure 8: Top 15 Feature Importances - Gradient Boosting Regressor.

The residual scatter plot (Figure 9) shows that prediction residuals are tightly clustered around zero across the full range of predicted consumption, with no systematic heteroscedasticity. The near-zero bias and homogeneous variance confirm that the model is well-calibrated and does not suffer from structural underfitting or overfitting.

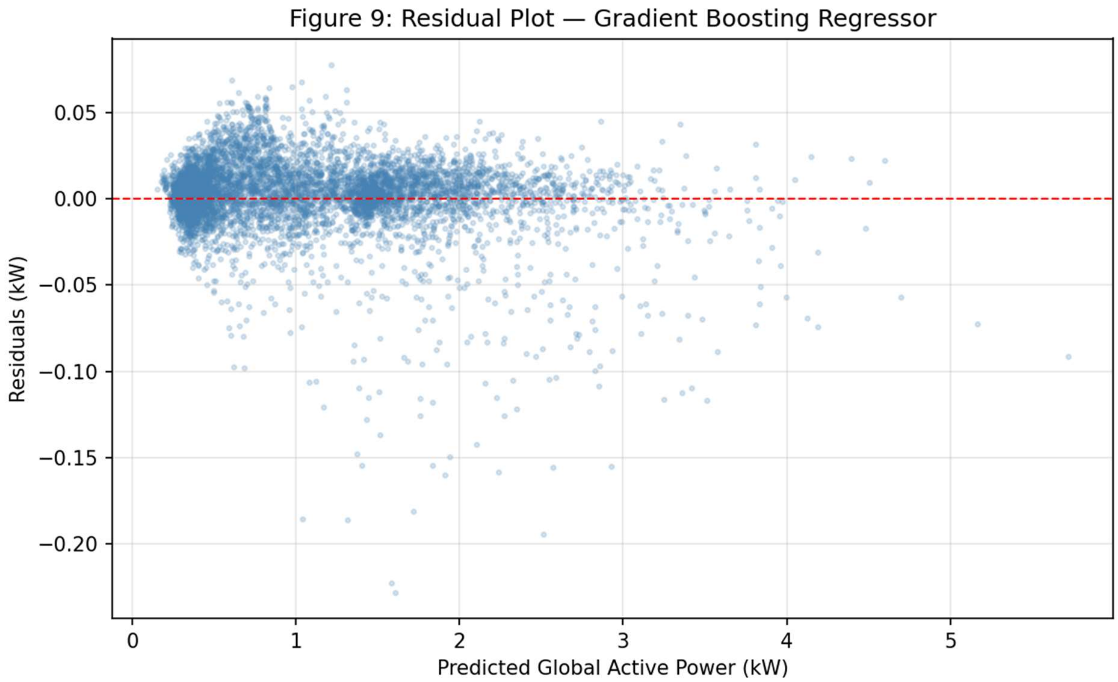

Figure 9: Residual Plot - Gradient Boosting Regressor.

9.4 Comparative Discussion

The superiority of ensemble methods over SARIMA can be attributed to three primary factors. First, Random Forest and Gradient Boosting exploit the full 20-feature multivariate input, including electrically informative attributes such as Global_intensity and sub-metering readings, whereas SARIMA operates on a univariate daily series. Second, Global_intensity effectively provides the models with near-direct algebraic access to the target through the physics of P = V x I. Third, the ensemble models represent non-linear interactions between temporal and electrical features through tree splits, which SARIMA cannot replicate. These findings align with Taylor and McSharry (2007), who reported rapid performance deterioration of ARIMA-class models beyond one-week horizons on residential load data, and with Amasyali and El-Gohary (2018), who found ensemble methods consistently outperform statistical baselines in residential forecasting.

10. Conclusion

This study implemented and evaluated a complete time-series forecasting pipeline for household electricity demand using the UCI Individual Household Electric Power Consumption dataset.

Three algorithms were assessed on a chronological 80/20 train/test split. The Gradient Boosting Regressor achieved the highest test-set accuracy (MAE = 0.0132 kW, RMSE = 0.0207 kW, R2 = 0.9992), marginally outperforming Random Forest (MAE = 0.0146 kW, R2 = 0.9990). Both ensemble methods substantially outperformed SARIMA(1,0,1)(1,0,1)[7] (R2 = -0.1941),

It should be noted that the near-perfect performance of ensemble models is largely driven by the inclusion of Global_intensity as a feature, which provides a direct physical proxy for the target variable. For purely forecasting applications where sub-metering and electrical attributes are unavailable at prediction time, a model architecture relying solely on lag and calendar features would provide a more practically relevant evaluation.

Several limitations should be acknowledged. The evaluation employed a single chronological train/test split; rolling-window or expanding-window cross-validation across multiple folds would yield more statistically robust generalisation estimates and reduce sensitivity to the specific split boundary chosen. No external meteorological data were incorporated into the feature set, despite ambient temperature and humidity being well-established drivers of residential electricity demand, particularly for heating and cooling loads (Hong and Fan, 2016). Additionally, the current pipeline does not address multi-step ahead forecasting, which is operationally important for grid operators who typically require demand predictions at horizons of six to twenty-four hours rather than a single one-step-ahead output.

Future work should pursue three directions. First, evaluating the pipeline without electrical attributes would provide a cleaner assessment of temporal feature predictive power. Second, incorporating weather covariates as exogenous inputs would address a key limitation of the current feature set. Third, deep learning architectures such as LSTM networks (Kong et al., 2019) or Temporal Fusion Transformers (Lim et al., 2021) should be benchmarked against the ensemble baselines to assess whether additional model complexity is justified in the absence of near-perfect electrical features.

Acknowledgement

Perplexity AI was used in the preparation of this report to support literature searching, identify potentially relevant academic sources, and assist with grammar correction and word-limit reduction during editing. All sources included in the reference list were checked by the author and cited as the original academic works rather than citing Perplexity AI as a source. The analysis, interpretation, discussion, and final written content submitted in this report are the author's own work and responsibility.

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